MSc Data Mining

Module outline: from raw data to a model you can defend

M Usman · Teaching Assistant, University of Huddersfield

Indicative module outline informed by canonical MSc curricula.

1 · Aims

Turn raw, messy data into patterns a business can trust

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What students can do by the end

- Frame a business question as a data-mining task and choose a method that fits it
- Ground every model in statistics: estimate, test, and quantify uncertainty
- Build and tune both **supervised** and **unsupervised** models
- Evaluate honestly: tell a real result from one that has fooled itself
- Reason about scale, data quality, class imbalance, and the ethics of acting on a model

36 hours

twelve one-hour lectures and twelve two-hour labs, paired one to one

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Lectures 1–4 · statistical foundations

1. **Data and descriptive statistics** · the KDD and CRISP-DM process; data types; central tendency, dispersion, skewness, correlation (Pearson against Spearman); graphical exploratory analysis; Anscombe's quartet
2. **Inferential statistics and hypothesis testing** · sampling distributions, the central limit theorem, confidence intervals; p-values, Type I and II error, power, effect size; t-test, χ^2, ANOVA; multiple comparisons (Bonferroni, Benjamini–Hochberg); the bootstrap
3. **Data preparation and feature engineering** · missing-data mechanisms (MCAR, MAR, MNAR) and imputation; scaling, encoding, discretisation, skew transforms; the data-leakage trap
4. **Evaluation methodology** · train, validation and test; k-fold and nested cross-validation; the bias–variance decomposition; precision, recall, F1, ROC-AUC, PR-AUC; RMSE, MAE, R^2; calibration

Lectures 5–8 · supervised learning

5. **Class imbalance** · why accuracy misleads on skewed targets; resampling and SMOTE, cost-sensitive learning, threshold-moving; PR-AUC and balanced accuracy; resample inside the fold only
6. **Linear regression** · ordinary least squares, the Gauss–Markov assumptions, coefficient inference; residual diagnostics and multicollinearity; bias–variance and regularisation (ridge, lasso, elastic net)
7. **Logistic regression and Naïve Bayes** · the logit link and maximum likelihood, decision boundaries, odds ratios; Bayes' theorem with conditional independence (Gaussian, multinomial, Bernoulli); Laplace smoothing
8. **Instance-based learning: k-NN** · distance metrics, the role of scaling, choosing k, weighted voting; the curse of dimensionality and approximate neighbour search

Lectures 9–12 · models at scale, and honesty

9. **Support vector machines** · maximum-margin classifiers, hard against soft margin (C, slack), the hinge loss and support vectors; the kernel trick (linear, polynomial, RBF); multiclass strategies
10. **Neural networks** · the perceptron and multilayer perceptron, activation functions, forward propagation and backpropagation, gradient descent (SGD, momentum, Adam); regularisation (dropout, L2, early stopping); convolutional and recurrent networks as further study
11. **Clustering and dimensionality reduction** · k-means (k-means++, choosing k by elbow and silhouette), hierarchical clustering, DBSCAN; PCA by singular-value decomposition, explained variance; t-SNE and UMAP for visualisation
12. **Association rules, ethics and synthesis** · frequent-itemset mining (support, confidence, lift), Apriori and anti-monotonicity, FP-growth; interpretability, fairness, privacy; the pipeline end to end

Labs 1–4 · statistics and evaluation in code

1. **Descriptive statistics and EDA** · summary statistics and graphical analysis on UCI Adult; reproduce Anscombe's quartet (pandas, seaborn)
2. **Statistical inference** · t-tests, χ^2, ANOVA and bootstrap confidence intervals on Palmer Penguins; Benjamini–Hochberg correction (SciPy, statsmodels)
3. **Leakage-safe preparation** · build a scikit-learn Pipeline and ColumnTransformer on Ames Housing; demonstrate the train-test contamination error, then correct it
4. **Evaluation and cross-validation** · stratified and nested cross-validation on Breast Cancer Wisconsin; ROC and precision–recall curves; learning and validation curves

Labs 5–8 · core algorithms by hand

5. **Class imbalance** · baseline against class weighting, SMOTE and threshold-moving on a fraud dataset, judged by PR-AUC and recall at fixed precision (imbalanced-learn)
6. **Regression and regularisation** · OLS, ridge, lasso and elastic net on Ames Housing; residual diagnostics and VIF; an R glmnet cross-check
7. **Logistic regression and Naïve Bayes** · text classification on the SMS Spam collection; interpret odds ratios; baseline comparison
8. **k-NN and the curse of dimensionality** · tune k and the distance metric on UCI Wine; measure the effect of feature scaling empirically

Labs 9–12 · scale, structure and a capstone

9. **Support vector machines** · grid-search C and the RBF γ on UCI Ionosphere; visualise decision boundaries; compare linear, polynomial and RBF kernels
10. **Neural networks** · train a multilayer perceptron on Fashion-MNIST; study learning rate, activation and dropout; plot training against validation loss
11. **Clustering and PCA** · k-means, hierarchical clustering and DBSCAN on customer data; choose k by silhouette; compare clusters in PCA space
12. **Capstone** · mine association rules with Apriori and FP-growth on UCI Online Retail, then assemble a full leakage-safe pipeline from exploration to honest evaluation

Tools and datasets

- **Python** (pandas, NumPy, SciPy, scikit-learn, statsmodels) and **R** for the hands-on labs
- Open, real-world datasets (UCI and similar) rather than toy examples
- Every result is reproduced from committed code; any number traces back to the analysis behind it

> A worked Python example for this module is public: heart-failure-clinical-ml-py.

How it is taught

- Each lecture pairs a method with the problem it was invented to solve, and the assumptions it rests on
- Each two-hour lab builds one stage of an end-to-end mining pipeline, from raw data to an evaluated model
- The recurring discipline: validate before you believe, and name the leakage before it flatters the score

Why it matters

A model is only as trustworthy as the evaluation behind it. The module puts honest evaluation first, before any result is acted on.

In summary

Core texts: Han, Kamber & Pei, Data Mining: Concepts and Techniques (3rd ed.) · James, Witten, Hastie & Tibshirani, An Introduction to Statistical Learning (2nd ed.) · Hastie, Tibshirani & Friedman, The Elements of Statistical Learning (2nd ed.) · Leskovec, Rajaraman & Ullman, Mining of Massive Datasets (3rd ed.)

M Usman · Teaching Assistant, University of Huddersfield · mtusman-ai.github.io/M.Usman

See the full portfolio

The work behind this deck sits on one page: the Insights series, the projects and the code that runs them, and the publications.

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